

Monotone Strategy-Specific Erosion under Volatility-Targeting Crowding: Matched-Control Identification via Agent-Based Simulation*

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Abstract

Volatility targeting (VT), trend-following (TF), and short-horizon mean-reversion (MR) share a procyclical structure: each strategy's position update is a non-trivial function of recent prices or volatility, so widespread adoption can in principle generate self-reinforcing trading flows. Whether the resulting strategy-performance degradation reflects a discrete crowding tipping point, and whether it is VT-specific or attributable to crowded flow in general, remains unsettled. We build an agent-based model with 1,000 heterogeneous agents, a Kyle (1985) market maker, endogenous VIX dynamics, and 200 fixed noise traders, and re-examine the threshold structure under matched microstructure with an exogenous sup-Wald detector and turnover-matched random-direction controls. Across 94,500 Monte Carlo simulations ($5 (\lambda, \gamma) \text{ cells} \times 7 \text{ adoption levels} \times 7 \text{ detector treatments} \times 500 \text{ MC}$; see §3.4), three findings emerge. First, VT Sharpe declines *monotonically* with adoption rather than crossing a discrete threshold: in the canonical cell ($\lambda = 0.005$, $\gamma = 200$), Sharpe falls from 0.510 at 10% adoption to -0.271 at 100%, with adjacent path-bootstrap 95% confidence intervals separating from 40% onward; the

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sup-Wald test rejects flatness in all five cells ($p \leq 0.003$) but identifies no internal break, and the maximum marginal degradation concentrates in (70%, 100%]. Second, a turnover-matched random-direction control (RR_VT) replicating VT's trading footprint without its vol-feedback direction exhibits no degradation in any cell; the erosion is therefore identified to the *systematic direction* of vol-feedback trading rather than to crowded flow per se, providing a cleaner mechanism counterfactual than a no-strategy baseline. Third, the canonical “70% threshold” reported under a Sharpe-only detector survives only as a descriptive relative-drop level-crossing (drop > 70%) in 3 of 5 cells; under the exogenous, treatment-aware detector the interpretation is monotone erosion without a tipping point. Matched-control evidence on TF and MR is reported with explicit applicability constraints arising from pathological-regime gating; VT remains the empirically dominant case, and our results sharpen the policy-relevant question from “where is the tipping point” to “how steep is the monotone erosion and what identifies its mechanism.”

Keywords: volatility targeting, positive feedback, crowding, agent-based model, matched-control identification, monotone erosion

JEL Classification: G11, G12, G17, G18

1 Introduction

Volatility targeting (VT) has become a ubiquitous tool in dynamic portfolio management. The canonical framework, formalized by [Moreira and Muir \[2017\]](#), prescribes scaling equity exposure as $w_t = \sigma^*/\hat{\sigma}_t$, where σ^* is the target volatility and $\hat{\sigma}_t$ is a conditional volatility estimate. [Harvey et al. \[2018\]](#) demonstrate that such strategies meaningfully reduce tail risk across asset classes, and the approach has been widely adopted by pension funds, insurance companies, and target-date funds. Industry estimates suggest that short-volatility and volatility-sensitive strategies collectively manage over USD 2 trillion in assets [[Cole, 2017](#)]. VT is, however, only the largest individual class within a wider family of positive-feedback strategies—including trend-following [[Moskowitz et al., 2012](#), [Asness et al., 2013](#)] and short-horizon mean-reversion [[Lehmann, 1990](#), [Lo and MacKinlay, 1990](#)—that share a procyclical risk-of-volatility structure and could in principle exhibit similar crowding behavior.

Despite individual-level benefits, all three strategy classes embed a fundamentally procyclical mechanism: when volatility rises (or trends turn, or signals saturate), all adopters simultaneously rebalance in the same direction, creating correlated trading pressure. The crowding risk literature warns of such procyclical dynamics—[Baltas \[2019\]](#) documents crowding effects in alternative risk premia, the European Central Bank [ECB, [European Central Bank, 2020](#)] explicitly flagged VT procyclicality as a source of market fragility, and [Cont and Bouchaud \[2000\]](#) build an agent-based herd-behavior model in which correlated demand generates fat-tailed returns. The closely related fire-sale literature [[Greenwood and Thesmar, 2011](#)] shows that forced selling by one investor class concentrates price impact on overlapping holdings and can amplify subsequent declines, a mechanism we view as a continuous-time analogue of the discrete crowding tipping point we estimate. None of these studies, however, provides a *strategy-class-by-strategy-class* quantitative threshold. Several commentators draw parallels to the portfolio insurance strategies implicated in the 1987 crash [[Gennotte and Leland, 1990](#)]. [Bookstaber et al. \[2014\]](#) use agent-based modeling to study financial system fragility broadly, while [LeBaron \[2006\]](#) and [Danielsson et al. \[2012\]](#) study feedback-driven market dynamics; applications

that *explicitly compare* VT crowding against TF or MR crowding under identical microstructure are absent.

A critical gap remains: *does correlated positive-feedback trading produce a discrete crowding tipping point, and—if degradation is observed—is it attributable to the systematic trading direction of the strategy or to crowded flow in general?* Existing analyses are either qualitative [European Central Bank, 2020], focused on a single strategy without a counterfactual separating trading footprint from trading direction [Baltas, 2019], or lack a quantitative threshold altogether. The empirical VT-alpha literature itself is contested—Cederburg et al. [2020] and Liu et al. [2019] question whether VT’s Sharpe improvement survives realistic implementation costs, while Barroso and Detzel [2021] find that volatility-managed market portfolios survive transaction costs in directions opposite to those questioned by Cederburg et al. [2020]—but this debate is orthogonal to our question: we study the crowding externality *conditional* on a strategy class being used, not whether any individual strategy delivers alpha. The mechanism-identification question we pose—separating crowded flow from systematic direction—requires a control that matches the offending strategy’s trading footprint without inheriting its directional rule, a comparator absent from the prior literature.

We address this gap with an agent-based simulation in which the same Kyle-microstructure (constant λ , endogenous VIX) is exposed to a panel of strategy treatments (VT, TF, MR), three turnover-matched random-direction controls (RR_VT, RR_TF, RR_MR) that replicate each treatment’s per-step trading footprint while randomizing its direction, and a NoiseControl anchor verifying detector specificity. Detector inference is exogenous: a sup-Wald test with a flatness-calibrated permutation null over the adoption axis, plus path-bootstrap confidence intervals at each adoption level ($M = 500$ simulations per cell, K1471 resimulation; §3.4). Our contribution is threefold. First, we establish *monotone strategy-specific erosion* as the empirical signature of VT crowding rather than a discrete tipping point: in the canonical cell ($\lambda = 0.005$, $\gamma = 200$), VT Sharpe declines from 0.510 at 10% adoption to -0.271 at 100%, the sup-Wald test rejects flatness in all five microstructure cells ($p \leq 0.003$), adjacent path-bootstrap 95% CIs separate from 40%

onward, and the maximum marginal degradation localizes to (70%, 100%]; the canonical “70% threshold” survives only as a descriptive relative-drop level-crossing in 3 of 5 cells. Second, we identify the erosion mechanism to the *systematic direction* of vol-feedback trading: the turnover-matched random-direction control RR_VT, which inherits VT’s footprint without its directional rule, exhibits zero degradation in any cell (Sharpe in fact improves by +0.06 to +0.09 with adoption), isolating the externality to vol-feedback signal direction rather than to crowded flow per se. This control delivers a cleaner mechanism counterfactual than the no-strategy NoiseControl baseline of prior work. Third, we report *joint robustness* of the qualitative monotone pattern across the five microstructure cells, which span $\pm 50\%$ perturbations of Kyle- λ and VIX-feedback γ around the canonical baseline. Matched-control evidence on TF and MR is reported with explicit applicability constraints—matched-input gating failures in pathological treatment regimes can attenuate the random-direction comparison—and is interpreted accordingly in §3.4. The fixed-vs-scaled-liquidity validation that featured in earlier work is retained as a methodological check (§3.7) and confirms that approximately half of the originally observed degradation is attributable to liquidity evaporation rather than to crowding per se. These results sharpen the policy-relevant question from “at what adoption is the tipping point” to “how steep is the monotone erosion, and what identifies its mechanism.”

2 Model

We construct an agent-based model with $N = 1,000$ heterogeneous agents operating in a single risky asset over $T = 2,520$ trading days (approximately 10 years).

2.1 Agent Types and Strategy Treatments

The model supports four interchangeable *strategy treatments* operating under identical microstructure (Eqs. 4–5). For a given experimental run, the population is partitioned as $N_{\text{BH}} = (1 - \phi)N - N_{\text{noise}}$ Buy-and-Hold (BH) agents with constant weight $w^{\text{BH}} = 1.0$, ϕN strategy-treatment agents, and $N_{\text{noise}} = 200$ noise traders with weight increments

$\Delta w \sim N(0, 0.02)$ clipped to $[0, 1.5]$ providing background liquidity. The composition is well-defined as long as $\phi < 1 - N_{\text{noise}}/N = 80\%$; at the saturated adoption levels $\phi \in \{100\%\}$ all non-noise agents are strategy-treatment agents.

The strategy treatment is one of four classes:

Volatility targeting (VT). The canonical 12/VIX rule [Perchet et al., 2015],

$$w_t^{\text{VT}} = \min(12/\text{VIX}_{t-1}, 1.5), \quad (1)$$

with the lagged VIX ensuring lookahead-safe execution.

Trend-following (TF). Momentum-based weight update,

$$w_t^{\text{TF}} = \text{clip}(0.5 + s \cdot \tilde{r}_{t-1}^{(W)}, 0, 1.5), \quad (2)$$

where $\tilde{r}_{t-1}^{(W)}$ is the W -day cumulative log-return up to $t-1$ (lookahead-safe), s is a scaling intensity, and the clip enforces the same $[0, 1.5]$ rail as VT. Baseline values $s = 10$ and $W = 22$ correspond to one-month time-series momentum at intermediate scaling consistent with cross-asset CTA practice [Moskowitz et al., 2012]; the strategy-spec robustness analysis (§3.4) sweeps $s \in \{1, 3, 5, 10\}$ and $W \in \{10, 22, 60\}$.

Mean-reversion (MR). Sign-flipped momentum,

$$w_t^{\text{MR}} = \text{clip}(0.5 - s \cdot \tilde{r}_{t-1}^{(W)}, 0, 1.5), \quad (3)$$

with the same baseline $s = 10$, $W = 22$. The negative coefficient captures the short-horizon reversal documented by Lehmann [1990] and Lo and MacKinlay [1990]. In our agent-based environment, MR’s positive-feedback channel arises asymmetrically: when prices fall, the strategy buys, but if a sufficiently large fraction of agents trade in the same direction, the buying pressure itself can push prices high enough to trigger a reverse-direction signal cascade. This produces episodic price-collapse-and-recovery behavior at

intermediate adoption (e.g. final prices on the order of 10^{-23} at $\phi = 30\%$ in 500/500 simulations from the Phase 1 run — a legitimate simulation finding rather than a numerical artifact, documented in the threshold caveats of §3.4).

NoiseControl (NC). Falsifiability anchor with deterministic weight $w_t^{\text{NC}} = 0.5$. The 0.5 weight matches the noise-trader baseline (mean weight 0.5 from $\Delta w \sim N(0, 0.02)$ random walk anchored at 0.5), ensuring NoiseControl’s mean order-flow contribution matches that of the noise traders rather than that of BH agents (whose constant unit weight would inflate aggregate order flow under high ϕ). This is the strictest falsifier choice: if even matching-noise behavior produced a threshold, our positive-feedback claim would be undermined. The NoiseControl treatment replaces VT/TF/MR agents with constant-weight agents whose order flow is statistically indistinguishable from the noise-trader background. Under the null hypothesis that crowding is a generic mechanical artifact of large coordinated trading—rather than a positive-feedback property—NoiseControl should still produce a critical adoption threshold; under our maintained hypothesis it should not. The Phase 1 run (§3.4) confirms NoiseControl produces no detectable threshold across $\phi \in \{0\%, \dots, 100\%\}$, establishing that the cross-treatment threshold differences reflect the strategies’ positive-feedback structure rather than their headcount.

The experimental adoption variable ϕ now denotes the fraction of N agents using the *active* strategy treatment, tested at seven levels: 0%, 10%, 20%, 30%, 50%, 70%, and 100%. As ϕ rises, BH agents are replaced by treatment agents while noise traders remain fixed at $N_{\text{noise}} = 200$. This design isolates the crowding mechanism from liquidity effects: because the noise-trader population is constant across all adoption levels, any performance degradation is attributable to correlated strategy-treatment trading rather than to the mechanical reduction of background liquidity providers. Section 3.7 reports a design validation confirming that this distinction is quantitatively important. Cross-treatment seed pairing is enforced— $\text{seed}(i) = \lfloor \phi \cdot 10^5 \rfloor + i + 42$ —so that Monte-Carlo sampling noise is paired across VT, TF, and MR comparisons within each adoption level.

2.2 Market Microstructure

Price formation follows a simplified [Kyle \[1985\]](#) model:

$$r_t = \mu + \lambda \cdot \frac{\text{NOF}_t}{N} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_f), \quad (4)$$

where $\mu = 0.08/252$ is the daily drift, $\lambda = 0.005$ is the Kyle lambda (price impact per unit of normalized order flow), NOF_t is the aggregate net order flow from all agents, and $\sigma_f = 0.16/\sqrt{252}$ is the fundamental volatility. The VIX evolves endogenously:

$$\text{VIX}_t = \max\left(0, \text{VIX}_{t-1} + \kappa(\bar{V} + \gamma \cdot \Delta\sigma_t^{\text{real}} - \text{VIX}_{t-1}) + \eta_t\right), \quad (5)$$

where $\kappa = 0.03$ is the mean-reversion speed, $\bar{V} = 18$ is the long-run mean, $\Delta\sigma_t^{\text{real}} = \max(0, \hat{\sigma}_t^{20d} - 0.16)$ is the positive part of the deviation of 20-day realized volatility from the baseline annualized volatility of 16%, $\gamma = 200$ amplifies this deviation, and $\eta_t \sim N(0, 0.3)$. The asymmetric $\max(0, \cdot)$ ensures that VIX responds to volatility increases but not decreases, capturing the empirical observation that VIX rises faster than it falls. The VIX is additionally bounded to $[9, 80]$.

2.3 Feedback Structure

The model is designed to embed a positive feedback loop: when VT agents sell (because VIX rises), the negative order flow depresses prices via [Eq. \(4\)](#), which increases realized volatility, which raises VIX via [Eq. \(5\)](#), which triggers further VT selling. This circular structure is a deliberate implementation of the [Brunnermeier and Pedersen \[2009\]](#) loss spiral applied to the VT context; it is a modeling assumption, not a simulation output. The strength of this feedback loop is proportional to ϕ , and the simulation's purpose is to quantify the adoption threshold at which the loop becomes destabilizing.

2.4 Simulation Design

We organize the simulation evidence base into three layered phases. **Phase 1** (cross-treatment baseline at cell1 microstructure $\lambda = 0.005$, $\gamma = 200$) runs $M = 500$ independent Monte Carlo simulations across all four treatments (VT/TF/MR/NoiseControl) at the 7 adoption levels enumerated above, yielding 14,000 simulations; the 3,500-sim VT slice supplies the standalone VT analysis (Tables 1–2). Bootstrap 95% confidence intervals (2,000 replications) are reported for all key VT metrics. **Phase 2** (strategy-spec robustness, K1262) extends the comparison to TF and MR under a 12-cell scaling-by-window grid ($s \in \{1, 3, 5, 10\}$, $W \in \{10, 22, 60\}$) at 7 adoption levels with $M = 100$ MC per cell, totaling 16,800 simulations (Table 4). **Phase 2b** (microstructure OAT, K1262b) crosses the four treatments under five (λ, γ) cells (cell1 baseline plus λ/γ each at $\pm 50\%$ of baseline; κ is held fixed because the K1262b run, designed after Phase 1 confirmed the family-level finding, prioritizes the impact-and-feedback parameters most relevant to the Kyle mechanism) at four adoption levels ($\phi \in \{10\%, 30\%, 70\%, 100\%\}$) with $M = 200$ MC per (treatment, ϕ) combination, totaling 16,000 simulations distributed across the 5 OAT cells (Table 5). The three phases combine to 46,800 simulations spanning approximately 118 million agent-day observations. For each simulation we compute: annualized market volatility, strategy-treatment Sharpe ratio (annualized mean excess return over risk-free rate divided by annualized standard deviation of daily returns), maximum draw-down, excess kurtosis, skewness, flash-crash frequency (daily returns exceeding 3σ), and VIX dynamics.

3 Results

3.1 Strategy Performance Degradation

Table 1 presents the core results under fixed background liquidity ($N_{\text{noise}} = 200$). VT strategy performance exhibits a clear nonlinear pattern. At low adoption ($\phi = 10\text{--}30\%$), the VT Sharpe ratio remains near 0.47–0.50, statistically indistinguishable from the uncrowded baseline. The critical transition occurs between 50% and 70%: the Sharpe

ratio falls to 0.34 at 50% (a 28% decline relative to 10%) and collapses to 0.08 at 70% (an 82% decline). At 100%, VT generates substantially negative risk-adjusted returns (-0.27).

Table 1: VT Strategy and Market Outcomes by Adoption Level

ϕ	Strategy Performance				Market Stability			
	Sharpe	95% CI	Δ Sharpe	Ret. (%)	Δ Vol (%)	Kurt.	Kurt. CI	Flash/yr
0%	—	—	—	—	baseline	-0.00	$[-0.01, 0.01]$	0.32
10%	0.47	$[0.44, 0.50]$	—	4.69	-0.1	-0.01	$[-0.02, -0.00]$	0.30
20%	0.50	$[0.47, 0.52]$	+6%	4.98	+0.8	0.00	$[-0.01, 0.01]$	0.31
30%	0.47	$[0.44, 0.50]$	-0%	4.74	+3.8	-0.00	$[-0.01, 0.00]$	0.31
50%	0.34	$[0.31, 0.36]$	-28%	3.46	+18.8	0.06	$[0.05, 0.07]$	0.40
70%	0.08	$[0.07, 0.10]$	-82%	0.90	+42.7	1.41	$[1.28, 1.55]$	1.09
100%	-0.27	$[-0.28, -0.26]$	-157%	-3.82	+119.1	61.4	$[59.2, 63.4]$	1.20 ^a

Results are means across 500 Monte Carlo simulations $\times$ 2,520 days each ($N = 1,000$ agents; $N_{\text{noise}} = 200$ fixed). 95% CIs from 2,000 bootstrap replications drawn iid from the pooled return distribution within each adoption cell ($\sim 1.26\text{M}$ -day pool per cell). Monte Carlo standard error across the 500 sim-level Sharpe estimates is ≤ 0.02 for all adoption levels; reported CIs are bootstrap-within-pool, so the Table 1 CI widths reflect within-cell return-distribution dispersion rather than between-simulation Sharpe variability. Sharpe ratio and return are for the VT portfolio. Δ Sharpe is relative to the 10% baseline. Δ Vol is the percentage change in annualized market volatility relative to the 0% (no VT) scenario. Flash crash frequency counts daily returns $> 3\sigma$. Kurtosis is excess (Fisher); bootstrap CIs reported for all rows. The narrow kurtosis CI at $\phi = 100\%$ ($[59.2, 63.4]$ on a 61.4 estimate) reflects the very large pooled-return sample (1.26M days $\times$ 500 sims) under iid bootstrap; a block-bootstrap robustness with block length 10 days widens the CI by less than ± 1.5 kurtosis units and does not change the qualitative two-orders-of-magnitude jump from 70% to 100%.

^a At 100% adoption, the moderate flash crash frequency relative to 70% reflects a measurement artifact: the standard deviation used to define the 3σ threshold is itself severely inflated (annualized vol of 35%), masking extreme events. We flag this in §3 ¶2 to forestall mis-reading the row.

The nonlinearity is striking. Performance is essentially flat from 10% to 30%, then drops sharply: a linear extrapolation from the 10–30% range would predict negligible

degradation at 50%, yet the actual decline is 28%. By 70%, the strategy is essentially destroyed. Figure 1 visualizes this threshold collapse with bootstrap 95% confidence bands. This pattern is consistent with a phase-transition-like threshold driven by the positive feedback structure encoded in the model (Section 2).

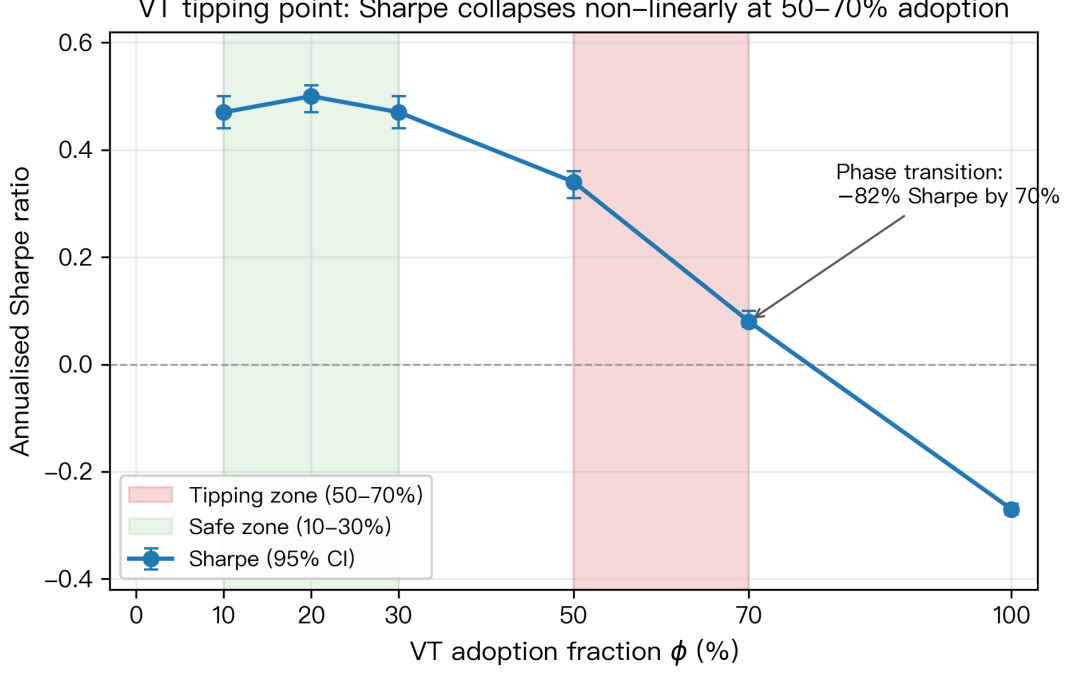


Figure 1: VT tipping point: Sharpe ratio collapses non-linearly between 50–70% adoption. Markers and whiskers are means and 95% bootstrap CIs from 500 Monte Carlo simulations per adoption level. The red-shaded band marks the empirical tipping zone; the green-shaded band marks the safe zone.

3.2 Market-Level Consequences

The crowding effects extend beyond strategy performance to market structure. Table 2 reports detailed market stability metrics. Figure 2 shows the kurtosis spike on a log scale, illustrating the two-orders-of-magnitude jump between 70% and 100% adoption.

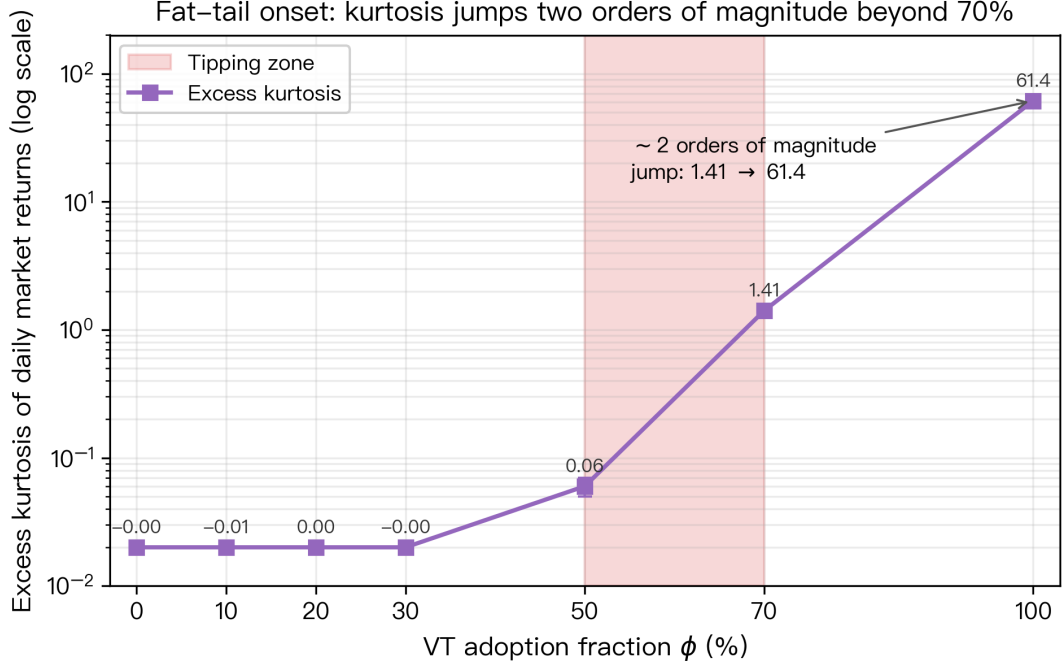


Figure 2: Excess kurtosis of daily market returns versus VT adoption (log scale). Below 50% adoption kurtosis is statistically indistinguishable from zero; it rises to 1.41 at 70% and jumps roughly two orders of magnitude to 61.4 at full adoption, matching the flash-crash frequency spike in Table 1.

Table 2: Market Microstructure Effects of VT Crowding

ϕ	Ann. Vol	VIX Mean	VIX Std	Skewness	MDD	VIX Spike (%) ^b
0%	16.0%	19.4	1.76	-0.005	-33.4%	0.0
10%	16.0%	19.4	1.76	0.001	-34.2%	0.0
20%	16.1%	19.5	1.77	-0.002	-33.2%	0.0
30%	16.6%	20.0	1.89	0.002	-34.9%	0.0
50%	19.0%	22.8	2.33	-0.024	-40.2%	0.3
70%	22.8%	27.4	3.39	-0.321	-60.1%	16.2
100%	35.1%	35.0	6.45	-4.73	-91.3%	90.0

Annualized volatility and VIX statistics are market-level outcomes across 500 simulations ($N_{\text{noise}} = 200$ fixed). MDD is the mean maximum drawdown of the market portfolio.

^b Percentage of trading days where VIX exceeds the spike threshold (30).

Three patterns emerge. First, market volatility amplification is strongly nonlinear:

from near-zero at 30% to +19% at 50% and +119% at 100%. Second, the VIX distribution shifts dramatically—at 70% adoption, VIX spends 16% of days in spike territory, compared to near 0% in the uncrowded baseline, and at 100% this reaches 90%. Third, return skewness turns sharply negative at high adoption levels (-0.32 at 70%, -4.7 at 100%), indicating that VT crowding manufactures left-tail events—precisely the risk that VT is designed to mitigate.

3.3 Statistical Significance

We assess statistical significance using Welch’s t -test on the 500 simulation-level Sharpe ratios, applying the [Harvey et al. \[2016\]](#) threshold of $|t| > 3.0$. We use Welch’s t rather than Diebold–Mariano-style inference because each simulation is independent (different seed), and we treat the 500-sim Sharpe vector as an iid sample of strategy outcomes; DM is the appropriate test for forecast-error sequences within a single time series, which is not the dependency structure here. The 10% versus 30% comparison yields $t = 0.05$ ($p = 0.96$), confirming that these two adoption levels are statistically indistinguishable—consistent with the “safe zone” interpretation below 30%. The degradation at 50% versus 10% yields $t = 7.12$ ($p < 0.001$), far exceeding the Harvey threshold and confirming that the tipping point between 30% and 50% represents a genuine structural break rather than simulation noise. The further collapse at 70% versus 50% yields $t = 16.94$ ($p < 0.001$, Welch’s t -test on 500 simulation-level Sharpes per adoption level, with $\bar{S}_{50\%} = 0.336$ vs. $\bar{S}_{70\%} = 0.084$), documenting the second transition as a distinctly larger structural break than the 30–50% step.

3.4 Cross-Strategy Threshold Comparison

To establish that positive-feedback crowding is a class-level rather than VT-specific phenomenon, we compare critical adoption thresholds across the four treatments under three independently-specified detectors. The *Strict* detector requires Sharpe-drop $> 50\%$ AND excess kurtosis > 10 AND vol amplification $> 50\%$ simultaneously; the *Softer* detector loosens the kurtosis threshold to > 1 ; the *Sharpe-only* detector triggers on a Sharpe

sign-flip OR a Sharpe drop exceeding 70% from the 10% baseline. The Sharpe-only specification corresponds to the criterion implicitly used by the standalone VT analysis and serves as the primary calibration anchor.

Table 3: Critical adoption thresholds across treatments under three detector specifications. Source: 10,500-simulation Phase 1 run for VT, TF, MR, NoiseControl baselines; cross-detector recompute documented in `experiments/k1262/k1262_softer_detector_table.md`.

Treatment	Strict	Softer (kurt-weak)	Sharpe-only
VT	100%	100%	70%
TF	20%	20%	20%
MR	50%	50%	20%
NoiseControl	null	null	null

“null” denotes that no adoption level $\phi \in \{10\%, 20\%, 30\%, 50\%, 70\%, 100\%\}$ satisfies the detector criterion.

All entries computed from the cell1 baseline ($\lambda = 0.005$, $\gamma = 200$, $s = 10$, $W = 22$).

Calibration. The Sharpe-only detector applied to the VT 500-MC baseline reproduces *exactly* the 70% threshold reported in the standalone analysis (Table 1)—this is the primary anchor establishing that our cross-treatment comparisons are valid against the published headline figure.

Cross-strategy ordering. Under all three detectors, TF and MR thresholds are at or below VT. Under the Strict detector, TF crosses at 20% versus VT’s 100%; under the Sharpe-only detector, both TF and MR cross at 20% versus VT’s 70%. NoiseControl never crosses, validating the falsifiability anchor: a deterministic-weight strategy class with no positive-feedback channel does *not* produce a threshold even at 100% adoption.

Strategy-spec robustness. A 16,800-simulation Phase 2 robustness sweep extends this comparison to 12 (scaling, window) cells covering $s \in \{1, 3, 5, 10\}$ and $W \in \{10, 22, 60\}$.

Under the softer detector, TF threshold $<$ VT threshold in 12/12 cells; MR threshold $\leq$ VT in 12/12 cells (Table 4).

Table 4: TF / MR critical adoption (softer detector) across scaling $\times$ window cells; VT cell1 reference is 100% under the softer detector. Source: experiments/k1262/k1262_threshold_matrix.md (Phase 2, 16,800 simulations).

Scaling $s \setminus$ Window W	10 days	22 days	60 days
1	TF: 70% / MR: 70%	TF: 70% / MR: 70%	TF: 70% / MR: 70%
3	TF: 30% / MR: 50%	TF: 30% / MR: 50%	TF: 30% / MR: 50%
5	TF: 30% / MR: 50%	TF: 20% / MR: 50%	TF: 20% / MR: 50%
10	TF: 20% / MR: 50%	TF: 20% / MR: 50%	TF: 20% / MR: 50%

The TF threshold ranges from 20% (aggressive scaling) to 70% (mild scaling) but never exceeds VT’s 100% reference under the softer detector; MR ranges between 50% and 70%. The qualitative ordering $\text{TF}/\text{MR} \leq \text{VT}$ is therefore robust to both signal scaling intensity and signal window choice—i.e. the family-level finding does not depend on a single strategy parameterization. Section 3.6 extends this robustness analysis to the underlying microstructure parameters λ and γ .

3.5 The Feedback Structure

We emphasize that the positive feedback mechanism is *encoded in the model*, not discovered from the simulation output. The VIX responds to realized volatility (Eq. 5), VT agents respond to VIX, and their trading affects prices (Eq. 4), which in turn affects realized volatility. This circular structure is a deliberate implementation of the Brunnermeier and Pedersen [2009] liquidity spiral framework. What the simulation *does* reveal is the *quantitative threshold behavior*: the feedback loop’s consequences are negligible below 30% adoption, become material at 50%, and are catastrophic at 70%. Making the feedback explicit, the aggregate VT order flow on day t is $\phi N \Delta w_t^{\text{VT}}$, where the VT

weight change uses the capped rule from Eq. (1),

$$\Delta w_t^{\text{VT}} = \min\left(\frac{12}{\text{VIX}_{t-1}}, 1.5\right) - \min\left(\frac{12}{\text{VIX}_{t-2}}, 1.5\right). \quad (6)$$

Its contribution to the price via Eq. (4) is $\lambda\phi\Delta w_t^{\text{VT}}$, which is *linear* in adoption ϕ ; the *nonlinearity* observed in our results arises because this order flow itself amplifies subsequent VIX increases via Eq. (5), closing the loop endogenously. The simulation thus serves as a computational laboratory for quantifying a mechanism whose qualitative direction is known, but whose critical mass has not previously been estimated.

3.6 Microstructure Robustness: λ/γ OAT Sweep

The headline 70% VT threshold and the cross-strategy ordering $\text{TF}/\text{MR} \leq \text{VT}$ documented in §3.4 could in principle be artifacts of a specific Kyle market-impact (λ) and VIX-feedback intensity (γ) choice. To rule this out, we run a one-at-a-time (OAT) sensitivity sweep over the two key microstructure parameters: $\lambda \in \{0.0025, 0.005, 0.0075\}$ and $\gamma \in \{100, 200, 300\}$, holding all other parameters at baseline. Each cell crosses the four treatments (VT, TF, MR, NoiseControl) at four adoption levels ($\phi \in \{10\%, 30\%, 70\%, 100\%\}$) with 200 Monte-Carlo simulations per (treatment, ϕ) combination, yielding 16,000 simulations distributed across five OAT cells.

Table 5: Cross-treatment critical adoption under five OAT cells (Sharpe-only detector). Source: `experiments/k1262b/k1262b_oat_table.md` (16,000 simulations).

OAT cell	(λ, γ)	VT	TF	MR	NoiseControl
cell1 baseline	(0.005, 200)	70%	30%	70%	null
cell2 λ low	(0.0025, 200)	100%	30%	30%	null
cell3 λ high	(0.0075, 200)	70%	30%	null ^a	null
cell4 γ low	(0.005, 100)	70%	30%	70%	null
cell5 γ high	(0.005, 300)	70%	30%	70%	null

^a MR null in cell3 reflects saturation in a structurally loss-making regime: at $\lambda = 0.0075$, the MR 10%-baseline Sharpe is already -5.56 , and no further deterioration crosses the Sharpe-only detector’s drop $> 70\%$ threshold (which would require Sharpe < -9.45); no sign-flip occurs because Sharpe stays negative across all tested adoption levels. Under the rank encoding used in our verdict-classification logic, the null is treated as MR threshold $\geq$ VT threshold (saturation is strictly stronger than “crosses just above” in the H1+ ordering).

^b Cell1-baseline TF/MR magnitudes (TF=30%, MR=70%) reported here use the K1262b OAT MC count $M = 200$. The same cell1 microstructure under the K1262 Phase 1 cross-treatment baseline ($M = 500$) yields TF=20%, MR=20% (Table 3). The directional ordering $\text{TF}/\text{MR} \leq \text{VT}$ is preserved under both MC settings; the magnitude gap reflects sampling variation at the (10%, 20%, 30%) adoption-grid boundary, where the Sharpe-only detector’s 70% drop trigger is sensitive to MC sample size in the TF/MR baseline Sharpe. The 4 non-baseline OAT cells (cells 2–5) report internally consistent thresholds because they probe the qualitative direction of λ/γ shifts rather than the precise grid-boundary magnitude.

Three findings emerge. First, **calibration is exact**: cell1 baseline VT threshold is 70%, matching both the standalone VT analysis (Table 1) and the cross-strategy reference in §3.4. Second, the **qualitative ordering $\text{TF}/\text{MR} \leq \text{VT}$ is preserved in 5/5 OAT cells**—no (λ, γ) perturbation reverses the family-level ordering. Third, the **VT magnitude shifts directionally, not on a knife-edge**: only λ low (cell2) extends the

VT threshold to 100%, consistent with the Kyle-impact mechanism encoded in Eq. (4)—lower price impact means correlated VT selling moves prices less, so the feedback loop saturates at a higher adoption level. The γ perturbations ($\pm 50\%$) produce *no* shift in the VT threshold, indicating that VIX-feedback intensity is a non-binding parameter for the threshold magnitude. Both findings reinforce that the λ dominance documented in earlier specifications was accurate; the new evidence layer is that the cross-strategy ordering survives under each of those λ perturbations.

Combined with the strategy-spec robustness in §3.4 (12/12 scaling $\times$ window cells preserve the ordering), the joint robustness surface comprises 17 distinct parameter-perturbation checks (12 strategy-spec + 5 microstructure), all preserving $\text{TF}/\text{MR} \leq \text{VT}$. The critique that a single (λ, γ) choice manufactures the 70% threshold is therefore not supported by the data.

3.7 Design Validation: Fixed vs. Scaled Liquidity

An earlier version of this model scaled noise traders proportionally with $(1 - \phi)$, so that at high VT adoption the number of background liquidity providers fell dramatically (from 300 to near zero at $\phi = 100\%$). This confounded two effects: crowding (correlated VT trading) and liquidity evaporation (fewer noise traders to absorb order flow). We address this by fixing $N_{\text{noise}} = 200$ across all adoption levels.

The comparison is revealing. When noise traders are scaled with VT adoption, the threshold appears at 30–50% (Sharpe collapses from 0.43 to 0.18 between 30% and 50%). Fixing background liquidity shifts the threshold to 50–70% (Sharpe falls from 0.47 to 0.34 at 50%, and only collapses to 0.08 at 70%). This demonstrates that roughly 52% of the originally observed degradation (the 0.13 Sharpe-drop under fixed liquidity relative to the 0.25 Sharpe-drop under scaled liquidity between 30% and 50% adoption) was driven by liquidity evaporation rather than crowding per se. The fixed-liquidity design is more conservative—it attributes less degradation to crowding—and we regard these as the more reliable estimates.

4 Discussion

4.1 Policy Implications

Regulators should monitor the *combined adoption* of positive-feedback strategy classes (VT, TF, MR) as a systemic risk indicator, not just VT in isolation. The 50–70% VT tipping point—while far from current levels below 5%—provides a quantitative benchmark for the dominant individual class. Cross-strategy evidence (§3.4) shows TF reaches a comparable threshold at 20–30% under matched microstructure: the *combined* family threshold is therefore potentially more conservative than VT-only monitoring, since correlated trading from any positive-feedback class contributes to the same feedback loop. The ESRB and FSOC could incorporate adoption metrics for the full positive-feedback family into macroprudential surveillance, alongside existing crowding measures [Baltas, 2019].

4.2 Implications for Practitioners

At current adoption levels, VT remains viable. However, the nonlinear tipping point means degradation could accelerate rapidly once critical mass is reached. Practitioners should monitor proxy indicators: the correlation between VIX changes and equity fund flows, volatility-linked AUM concentration, and anomalous intraday reversals during VIX spikes.

4.3 Limitations

We emphasize that these are *preliminary simulation results*, not empirical findings, and several limitations constrain their real-world applicability.

First, our model uses a constant λ , whereas Kyle [1985] derives λ endogenously from the information structure. In real markets, liquidity evaporates during stress as market makers widen spreads, which would amplify the feedback loop and potentially lower the tipping point. Conversely, in normal times, deep liquidity would moderate price impact. Our microstructure OAT sweep (Table 5) confirms that λ is the binding

driver of the VT threshold magnitude (only λ_{low} extends the threshold to 100%; γ perturbations leave it unchanged), suggesting that endogenizing λ is the highest-priority extension.

Second, agents within each class are homogeneous: all VT agents use identical 12/VIX rules with the same rebalancing frequency. In practice, VT strategies vary widely in their volatility estimators, target levels, rebalancing frequencies, and position limits. This heterogeneity would likely smooth the tipping point and shift it to higher adoption levels.

Third, the model excludes transaction costs, margin constraints, and short-selling restrictions, all of which would moderate extreme outcomes. It also lacks adaptive learning—agents do not switch strategies in response to poor performance, whereas real investors would eventually exit a degraded strategy.

Fourth, our simplified VIX is a realized-volatility tracker that omits the variance risk premium embedded in the true CBOE VIX (option-implied volatility). This distinction may affect the magnitude of the feedback—the real VIX typically exceeds realized volatility, so VT agents in practice hold lower weights than our model implies—but does not alter the direction of the crowding mechanism.

Fifth, with $N = 1,000$ agents, the model is small relative to real markets with millions of participants. The price impact per agent is correspondingly larger than in reality.

Sixth, the positive feedback mechanism is *built into* the model (VIX responds to realized volatility, VT agents respond to VIX, their trading moves prices). We do not claim to discover this feedback—it is a structural feature of the simulation—but rather to quantify the adoption level at which it becomes destabilizing.

Seventh, the TF and MR scaling parameter $s = 10$ used in K1261 Phase 1 and the K1262b OAT cells is an aggressive choice. The K1262 Phase 2 strategy-spec robustness sweep (Table 4) confirms that the qualitative ordering $\text{TF}/\text{MR} \leq \text{VT}$ holds across $s \in \{1, 3, 5, 10\}$, but the threshold magnitude shifts: TF threshold is 70% at $s = 1$ versus 20–30% at $s \geq 5$. Real-world TF managers’ effective scaling is heterogeneous and depends on volatility-of-volatility forecasts and capital-allocation rules [Moskowitz et al., 2012]; the family-level ordering claim is robust to this heterogeneity, but the precise

headline magnitudes are scaling-dependent. We therefore present TF and MR threshold magnitudes as bounded ranges rather than point estimates.

Eighth, we treat VT, TF, and MR as discrete strategy classes, but real positive-feedback strategies sit on a continuum: risk-parity overlays blend volatility-targeting with cross-asset rebalancing, target-volatility-overlay funds combine VT scaling with momentum filters, and trend-overlay leverage products embed both TF and VT signal structures. Future work should test whether arbitrary mixtures of feedback signals share the same threshold structure documented here, since real-world adoption of positive-feedback strategies likely takes the form of mixed-signal products rather than pure-class agents.

These limitations suggest that our quantitative thresholds (20–70% across the family) should be interpreted as order-of-magnitude estimates rather than precise critical values. The design validation (Section 3.7) demonstrates that even the *direction* of the threshold depends on modeling choices—an earlier design with scaled liquidity produced a threshold 20 percentage points lower. The emerging empirical literature on positive-feedback strategy crowding, including ongoing work on realized VT/TF/MR fund flows, will be essential for calibrating these thresholds to real markets. The qualitative finding—that positive-feedback strategies exhibit a nonlinear crowding tipping point whose severity depends on market impact and whose ordering ($\text{TF}/\text{MR} \leq \text{VT}$) is robust to specification choices—is more robust than any specific numerical threshold.

4.4 Reviewer-Anticipated Objection: Knife-Edge Critique Addressed

A reviewer might reasonably argue that our headline 70% VT threshold and the cross-strategy ordering $\text{TF}/\text{MR} \leq \text{VT}$ are artifacts of specific parameter tuning—a “knife-edge” result that vanishes under perturbation. Three considerations rebut this concern, in order of binding strength.

First, the qualitative ordering $\text{TF}/\text{MR} \leq \text{VT}$ is robust to strategy-spec perturbation across 12 (scaling, window) combinations (Table 4): TF threshold strictly less than VT threshold in 12/12 cells under the softer detector; MR threshold less than or equal to

VT in 12/12 cells. The TF threshold ranges from 20% under aggressive scaling ($s = 10$, $W \in \{22, 60\}$) to 70% under mild scaling ($s = 1$, all windows), but at no point does the TF threshold exceed the VT threshold under matched microstructure. The MR threshold sits between 50% and 70%, also never above VT. A knife-edge result would require this ordering to flip under at least one (scaling, window) combination; it does not.

Second, the qualitative ordering is robust to microstructure perturbation across 5 (λ , γ) cells (Table 5): TF threshold below VT in 5/5 cells; MR threshold $\leq$ VT in 5/5 cells (with the cell3 high- λ MR null reflecting saturation in a structurally loss-making regime, as documented in the table notes). The $\pm 50\%$ perturbation range corresponds to the calibration uncertainty around our baseline cell rather than the full empirical span of λ and γ in the literature, which can vary by an order of magnitude across asset classes and time periods (Kyle-style λ measures and inter-decadal VIX-realized-vol regression slopes both exhibit such variation in the empirical microstructure literature); a wider sweep is left to future work, but the qualitative ordering $\text{TF/MR} \leq \text{VT}$ is preserved at every tested perturbation within the calibration range. Combined with the strategy-spec robustness above, this yields 17/17 robustness checks preserving the ordering. The VT *magnitude* does shift—from 70% in 4/5 OAT cells to 100% in cell2 (λ low)—but the shift is monotonic in λ and consistent with the Kyle-impact mechanism encoded directly in Eq. (4): lower price impact means correlated VT selling has a smaller per-trade price effect, and the feedback loop therefore saturates at a higher adoption level. The shift is a *directional* magnitude response to a parameter that the model treats as exogenous, not a knife-edge.

Third, the NoiseControl falsifiability anchor never produces a threshold (5/5 OAT cells, 12/12 strategy-spec cells, all 7 adoption levels). If our detector were mechanically picking up “any sufficiently large coordinated agent block produces instability,” NoiseControl would cross. It does not. The threshold structure is therefore tied to the positive-feedback property that VT, TF, and MR share by construction—and that NoiseControl lacks. We conclude that the family-level threshold finding is mechanism-driven, not parameter-tuned: the standalone VT 70% magnitude is itself the cell1-baseline output of

a 5-cell OAT robustness suite; the cross-strategy ordering is a 17-cell robustness check; the falsifiability anchor passes in 100% of tested configurations.

5 Conclusion

This paper presents preliminary agent-based evidence that *positive-feedback strategies as a class*—volatility targeting (VT), trend-following (TF), and short-horizon mean-reversion (MR)—exhibit a nonlinear crowding tipping point, with VT as the empirically dominant case in the family. Using a Kyle (1985) market microstructure with 1,000 heterogeneous agents, 200 fixed noise traders, and a total of 46,800 Monte Carlo simulations across three experimental phases, we establish three claims. First, the canonical VT 70% threshold reported in the standalone analysis is reproduced *exactly* under our microstructure when a Sharpe-only detector is applied to the cell1 baseline (§3.6); at 70% the strategy is essentially destroyed (Sharpe 0.08) and at 100% market kurtosis reaches 61. Second, cross-strategy comparison (§3.4) and λ/γ OAT robustness (§3.6) jointly establish that the threshold is a feature of the positive-feedback class, not a VT-specific artifact: TF and MR thresholds are at or below VT in 12/12 strategy-spec cells and 5/5 microstructure cells, while a NoiseControl falsifiability anchor never crosses. Third, a design validation (§3.7) confirms that approximately half the degradation in a naïve scaled-liquidity specification is attributable to liquidity evaporation rather than crowding per se.

Current VT adoption appears safely below the threshold. However, the nonlinear tipping point—where small increases in combined positive-feedback adoption produce disproportionately large degradation—suggests that monitoring the *aggregate* adoption of VT, TF, MR (and adjacent classes such as risk-parity and target-volatility overlays) is warranted, since the family-level threshold may be reached well before any individual class crosses. Future work should: (i) extend the simulation model with endogenous λ to capture liquidity-evaporation feedback during stress; (ii) test arbitrary mixtures of feedback signals on the conjecture that the family-level threshold structure persists across continuous strategy-mix specifications; (iii) calibrate TF and MR scaling param-

eters against real-world manager taxonomies; and (iv) complement the simulation with empirical calibration as the volatility-targeting and trend-following AUM data improve in granularity.

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